



Introduction to **Web Programming**

Presented by
Nguyễn Đức Thái
Ho Chi Minh City University of Technology

Outline

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Creating a Website

- A website is a **collection** of related web **pages** that are normally **stored** on a single web server computer.
- A **web server**: *refer to* the web page-accessing **software** that runs on the web server *computer*.
- To create a website, you'll need these things:
 - (1) a text **editor**,
 - (2) an upload/publishing **tool**,
 - (3) a web **hosting service**,
 - (4) a **browser**.

Text Editor

- *Plain text editor*: such as Microsoft's Notepad.
- There are many *different text editors*, with varying degrees of functionality.
- Another feature common to all web authoring tools is **WYSIWYG**, pronounced “wizeewig”.
- It stands for “**W**hat **Y**ou **S**ee **I**s **W**hat **Y**ou **G**et”.
- WYSIWYG means that as you're editing your text, you *can see* what your text will look like after it's eventually uploaded to a website.
- Good Text Editors: **Dreamweaver**, **Sublime Text 3**,
...

WYSIWYG Editor

B*I*~~S~~{ }U

☰

☷

”

🔗

🔄

Normal

↶

↷

Hello World !

☒ HTML ☐ Markdown ☐ Editor is read-only

```
<p>Hello World !</p>
```

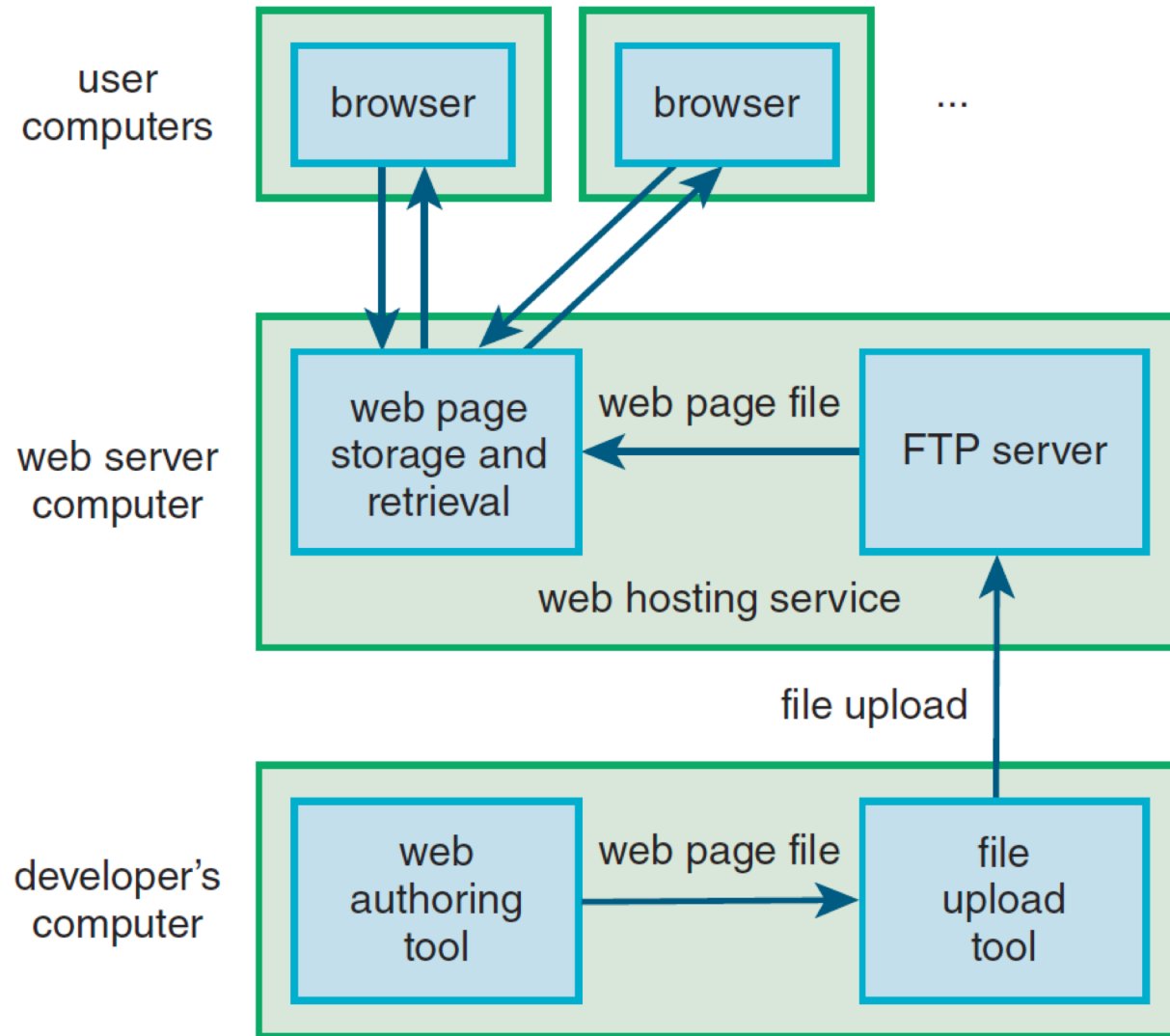
Upload / Publishing Tools

- **Publishing** means that you upload your web page to a web server computer so other users can access it on the Web.
- Some IDEs (**I**ntegrated **D**evelopment **E**nvironment), like Dreamweaver, provide *built-in* uploading capabilities, but other IDEs, like Visual Studio, do not.
- You could use a *separate* file upload *tool*, example **WinSCP**.

Web Hosting Service

- Need to have a *web server* **computer** on which to store the uploaded files.
- Web server computer needs to have a web hosting service in place.

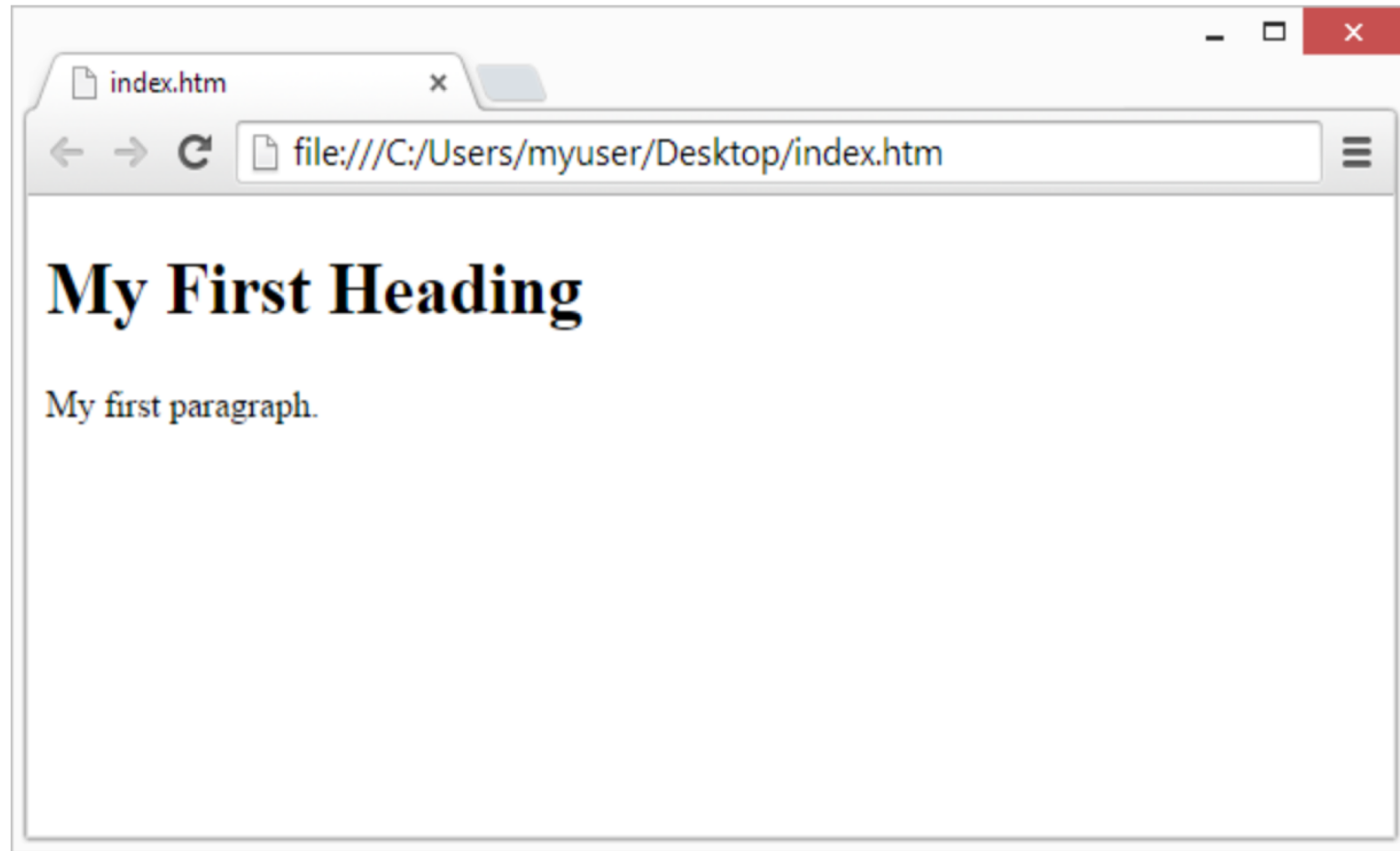
Web Hosting Service (2)



Browsers

- A **browser** is a *piece of software* that enables a user to retrieve and view a web page.
 - Google Chrome
 - Mozilla Firefox
 - Internet Explorer
 - Safari (for Mac devices)
 - Opera (for Android devices)
 - ...

Web Page Example



HTML

HTML = HyperText Markup Language



HTML Tags

<tagname>Content goes here...**</tagname>**

Start tag

End tag

Structural Elements

```
<!DOCTYPE html>  
<html>  
  <head>...</head>  
  <body>...</body>  
</html>
```

HTML Structure

`<!DOCTYPE html>`

Declaration of HTML5 document

`<html>`

root element of an HTML page

`<head>`

meta information about the HTML page

`<title>Page Title</title>`

`</head>`

element defines a large heading

`<body>`

container for all the visible contents

`<h1>This is a Heading`

`<p>This is a paragraph.</p>`

`</body>`

`</html>`

element defines a paragraph

HTML is **Not** Case Sensitive

- HTML tags are not case sensitive: **<P>** means the same as **<p>**.
- The HTML standard **does not** require lowercase tags, but W3C recommends lowercase in HTML, and demands lowercase for stricter document types like XHTML

head Element

- The **head** element contains two types of elements—*meta* and *title*

title Element

- First element in the head section.
- The text between the <title> and </title> tags will *appear* in the *title bar* of the browser window.
- This title text is accessed when web pages are *bookmarked* and *printed*.

title Element (cont'd)

- Search engines, such as Google, use the title text to help **determine** *keyword relevance* and even **display** the *title text* on the results page of a search.
- A *descriptive title* that includes the *website* or *organization name* is a crucial component for establishing a brand or presence on the Web.

meta Element

- The *meta* elements provide information *about the web page*.
- There are many *different types* of meta elements—some you should always include, but most are just *optional*.
- The meta element: NO *end tag*.

HTML Attributes

- HTML attributes provide *additional information* about HTML elements.
- All HTML elements *can have* attributes
- Attributes are *always* specified *in the start tag*
- Attributes usually come in *name/value pairs* like: name="value"

HTML Attributes (cont'd)

■ Element

- **Container elements:** provide information between their start and end tags
- **Void elements:** have no end tags, so they can't provide information that way

■ Void elements provide information *using* **attributes**.

- *For example*, charset is an attribute for a meta element:

```
<meta charset="utf-8">
```

■ Most attributes *have a value* assigned to them.

meta *name* Element

- *Most* of the meta elements use the *name attribute* to specify the *type of* information that's being provided.
- Common values for the meta name attribute are *author*, *description*, and *keywords*.
 - *Example:*

```
<meta name="author" content="David Johnson ">
```

- The *name* and *content* attributes go *together*.

meta *name* Element (2)

- In the following examples, the name attribute uses the values “**description**” and “**keywords**”:

```
<meta name="description" content="Kansas City weather conditions">
```

```
<meta name="keywords" content="KC, weather, meteorology, forecast">
```

- The meta **description** element and also the meta **keywords** element help **web search engines** *find your web page*.

body Elements: hr, p, br, div

- `<hr>` ➔ horizontal line.
- `<p>...</p>` ➔ for a *group of words*, form a paragraph
*(Whenever enclosed text is greater than one line, you should put the **start** and end **tags** (p, div) on **separate lines** and indent the enclosed text)*
-
- `<br>` ➔ break line
- `<div>...</div>` ➔ division.

CSS Preview

```
<head>|
```

-
-
-

CSS property

CSS value

```
<style>
```

```
  h1 {text-align: center;}
```

```
  hr {width: 75%;}
```

```
</style>
```

```
</head>
```

CSS property

CSS value

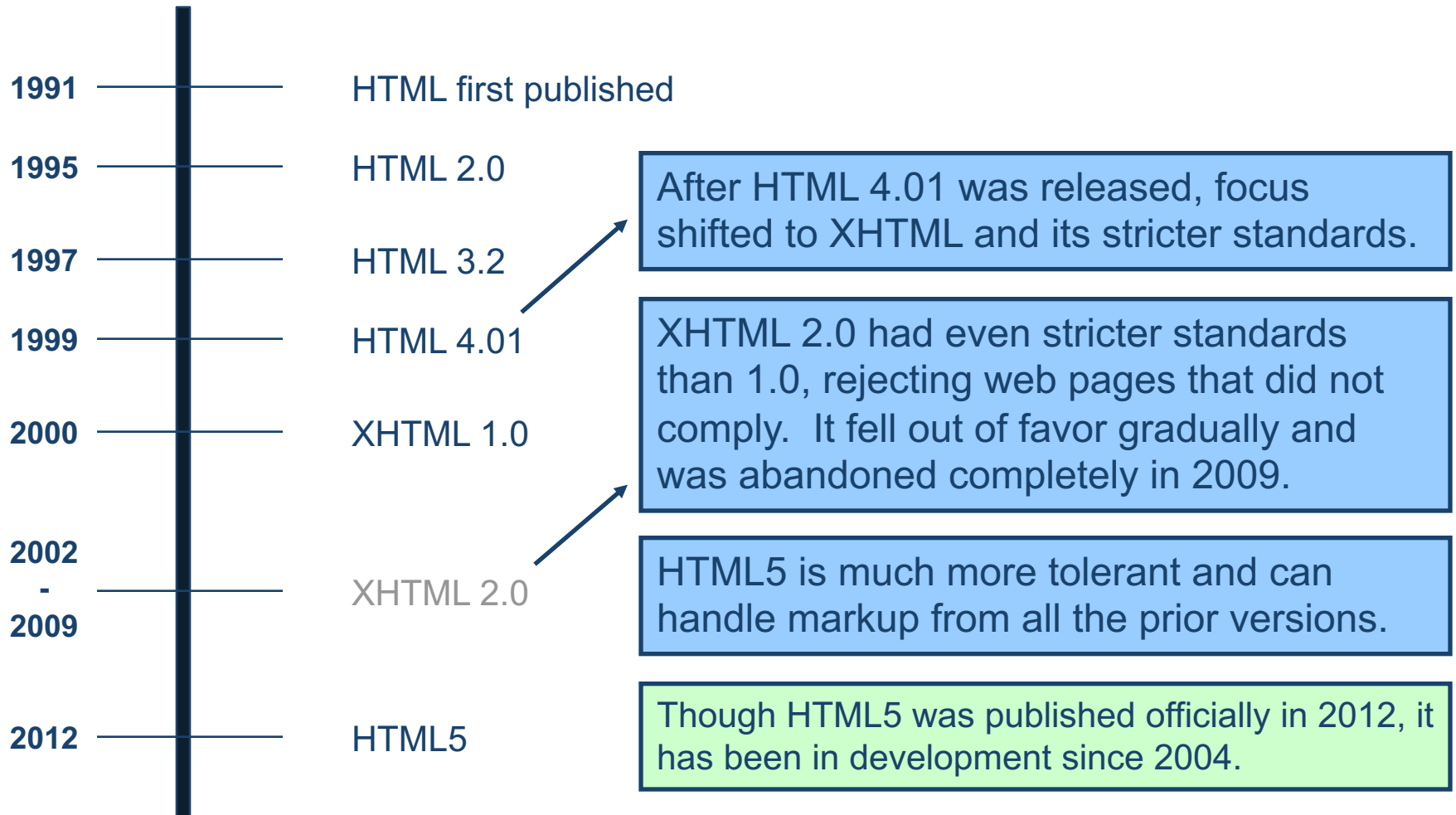
```
<body>
```

```
<h1>Kansas City Weather</h1>
```

```
<hr>
```

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History of HTML



References

- W3 Schools – [w3schools.com](https://www.w3schools.com)
- ***Web Programming with HTML5, CSS, and JavaScript*** – John Dean – Jones & Bartlett Publishers, 2019.